

HAYWARD, CA, USA

WMO: 725850

Lat: 37.654N

Lon: 122.115W

Elev: 43

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 93228

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.0	(c) 39.1	(d) 18.4	(e) 13.9	(f) 53.0	(g) 23.7	(h) 18.0	(i) 50.6	(j) 20.9	(k) 52.9	(l) 18.8	(m) 53.8	(n) 3.3	(o) 60	(p) 0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	17.4	87.4	65.0	82.7	64.0	78.8	63.1	67.5	82.1	65.9	79.1	64.5	76.1	10.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	84.1	71.1	61.0	80.2	69.4	59.8	76.7	68.5	31.8	82.0	30.6	79.1	29.5	76.2	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 17.6	(c) 16.1	(e) 33.3	(f) 96.4	(g) 2.0	(h) 4.6	(i) 31.9	(j) 99.7	(k) 30.8	(l) 102.4	(m) 29.7	(n) 105.1	(o) 28.2	(p) 108.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	51.0	53.0	55.8	57.5	60.6	63.9	65.9	66.4	66.7	63.0	56.1	51.1
	DBStd	7.28	4.22	4.33	4.78	4.84	4.63	5.01	4.00	3.68	5.12	4.59	4.51	4.56
	HDD50	110	39	18	5	1	0	0	0	0	0	0	6	41
	HDD65	2414	435	337	289	234	156	78	32	23	32	96	270	432
	CDD50	3492	69	101	185	226	329	416	494	508	501	402	188	73
	CDD65	319	0	0	3	8	20	45	60	66	83	33	1	0
	CDH74	2168	1	4	26	91	195	368	345	316	576	235	11	0
	CDH80	693	0	0	3	21	64	133	101	96	218	56	1	0
Wind	WSAvg	6.9	5.1	6.3	7.1	8.3	8.7	8.4	8.2	7.6	6.6	5.6	5.0	5.5
Precipitation	PrecAvg	19.6	3.9	3.6	2.7	1.3	0.7	0.2	0.0	0.0	0.1	1.0	2.2	4.0
	PrecMax	33.8	9.4	12.7	7.6	4.1	2.6	1.6	0.1	0.8	0.4	3.7	6.6	9.7
	PrecMin	7.8	0.3	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	5.6	2.6	2.5	2.2	1.0	0.7	0.4	0.0	0.2	0.1	1.0	1.6	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	72.3	77.4	82.8	87.3	91.1	89.0	89.7	92.6	86.1	74.9	66.8
		MCWB	51.9	56.4	60.1	62.6	63.3	66.3	67.6	66.4	65.9	61.2	58.5	55.0
	2%	DB	63.3	66.6	71.9	75.6	79.5	83.3	81.9	81.5	86.6	80.5	70.1	63.1
		MCWB	52.8	54.9	57.8	59.0	61.7	64.5	65.8	65.7	64.5	61.2	56.9	54.7
	5%	DB	60.6	63.4	67.6	70.0	72.9	77.5	77.2	76.6	80.6	75.7	66.6	60.9
		MCWB	53.1	54.3	56.2	57.6	59.8	63.0	64.4	64.3	63.6	59.8	56.2	54.0
	10%	DB	58.3	60.7	64.1	65.4	68.2	72.6	73.6	73.4	75.6	71.9	63.9	58.9
		MCWB	52.1	52.9	54.8	55.4	58.4	61.4	62.7	62.9	62.7	59.5	55.8	52.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	59.5	62.0	64.7	66.5	68.5	69.1	69.0	68.6	66.5	62.5	60.2
		MCDB	61.2	65.3	73.4	79.5	82.5	86.1	85.1	83.5	84.3	76.9	69.2	63.8
	2%	WB	56.1	57.4	59.5	60.9	63.2	65.9	66.4	66.3	66.6	64.1	60.4	58.0
		MCDB	59.7	63.8	68.5	73.3	75.8	80.7	80.3	78.8	80.2	73.9	66.3	60.8
	5%	WB	54.7	55.9	57.7	58.6	60.6	63.8	64.6	64.8	65.0	62.4	59.0	56.2
		MCDB	58.4	61.4	65.0	67.8	71.1	76.4	76.2	75.6	77.6	71.4	64.3	59.3
	10%	WB	53.2	54.4	56.2	56.5	58.8	61.7	63.1	63.4	63.6	60.9	57.3	54.2
		MCDB	57.1	59.2	62.6	64.0	67.4	72.3	72.9	72.7	74.1	69.2	62.4	57.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	14.4	15.1	15.2	14.8	16.0	15.9	15.2	17.4	17.1	15.2	13.9
		MCDBR	17.3	19.5	20.9	24.1	24.9	25.8	23.0	21.7	27.3	24.7	19.3	15.6
		MCWBR	10.2	11.0	11.4	12.4	12.0	11.8	10.3	10.0	12.5	12.5	10.6	9.1
	5% WB	MCDBR	12.2	15.4	17.5	21.1	21.3	23.1	21.3	20.1	22.9	19.8	15.0	10.3
		MCWBR	8.3	9.6	10.2	11.5	11.0	11.0	9.7	9.5	10.9	10.9	9.8	7.3
Clear-Sky Solar Irradiance	taub	0.324	0.331	0.342	0.360	0.367	0.361	0.368	0.349	0.350	0.338	0.325	0.326	
	taud	2.543	2.524	2.480	2.399	2.405	2.426	2.411	2.459	2.445	2.520	2.541	2.544	
	Ebn at Noon	274	287	293	292	290	290	287	291	285	279	271	265	
	Edh at Noon	24	28	32	37	37	36	37	34	33	28	24	23	
All-Sky Solar Radiation	RadAvg	757	1059	1499	1955	2263	2480	2356	2105	1775	1307	898	667	
	RadStd	106	127	134	157	143	102	75	85	66	81	56	96	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.65	N/A	-1.13	-1.64	N/A	-313	+374	N/A
(47) Regional (44 neighbors)	N/A	N/A	-2.11	N/A	N/A	N/A	N/A	-198	+243	N/A

Nomenclature: See separate page